Introduction to Bio-Inspired Design in Engineering

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Learning Objectives

- Introduction to Bio-Inspired Design
- Rawling Classification
- Evolution of Bio-Inspired Design
- Web Resources of Bio-Inspired Design
- Design Forms

Bio-Inspired Design

- It is the process of developing concepts, approaches and technologies that build and control the way nature does.

- It offers potentially transformative solutions to these problems.

- What does NOT come under Bio-Inspired Design:
  - Aesthetic-only designs
  - Pure human-invented technologies
  - Conventional engineering optimizations

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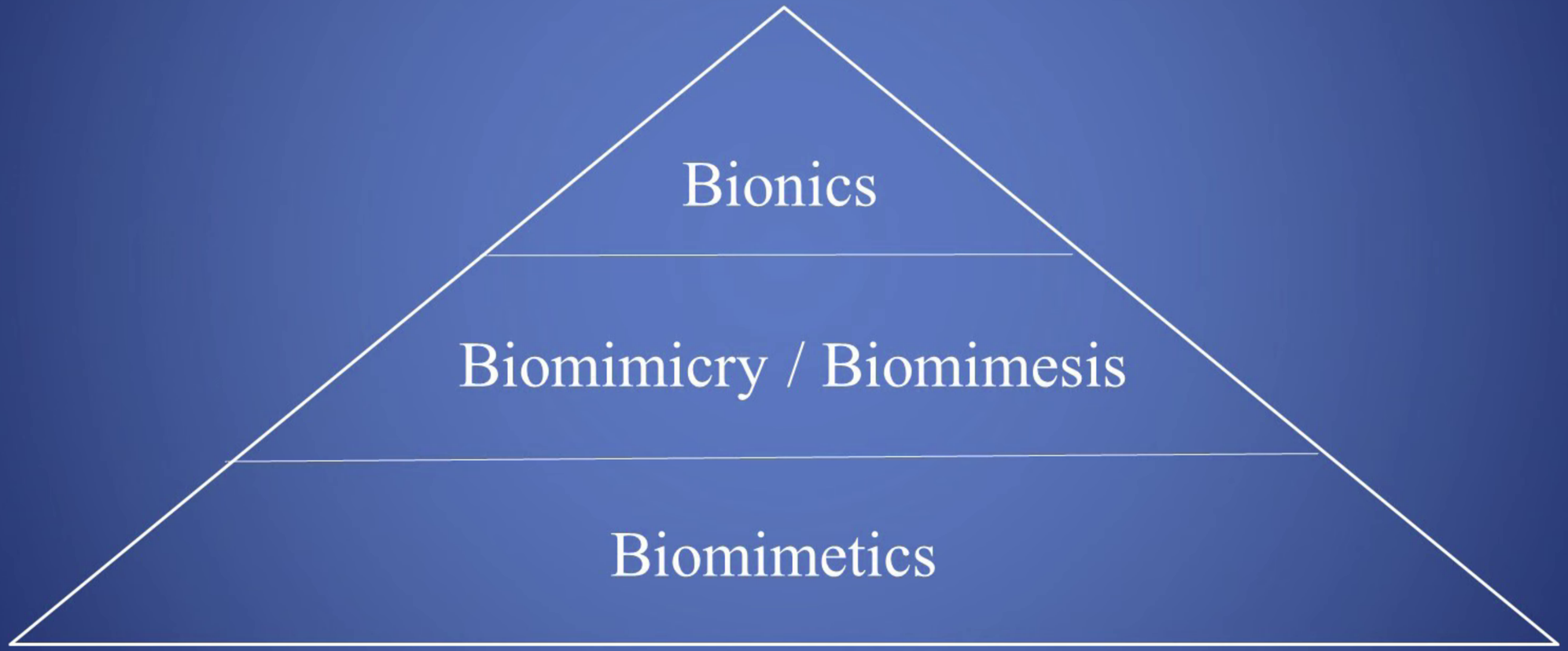
Bio-Inspired Design – Rawling Classification



Bio-Inspired Design

- **Biomimetics:**

  creative process between biology and technology, aimed at solving the anthropospheric problems like pollution, climate change etc., through abstraction and knowledge transfer from biological models.

- **Biomimicry / Biomimesis:**

  deals with the philosophy that takes-up challenges related to resilience (social, environmental and economic ones), by being inspired from living organisms, particularly on an organizational level.

- **Bionics:**

  technical discipline that seeks to replicate, increase or replace biological functions by their electronic and/or mechanical equivalents.

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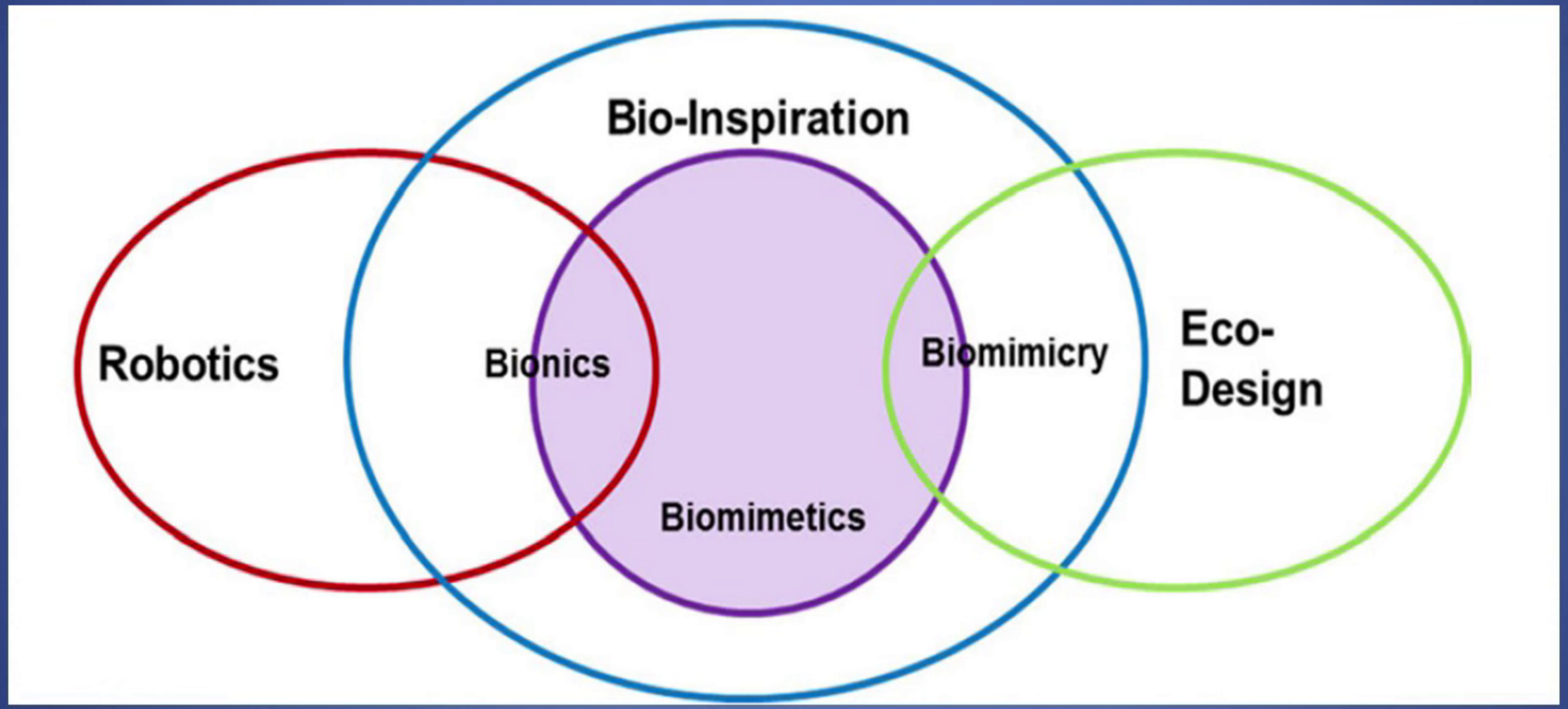
Bio-Inspired Design



Bio-Inspired Design

Examples:

- Biomimetics:
  - Shark-skin inspired swimsuits
  - Lotus-effect paints
- Biomimicry:
  - Termite Mounds inspired Urban Planning
  - Natural Ecosystem inspired Circular Economy models
- Bionics:
  - Bionic prosthetic arms
  - Cochlear implants.

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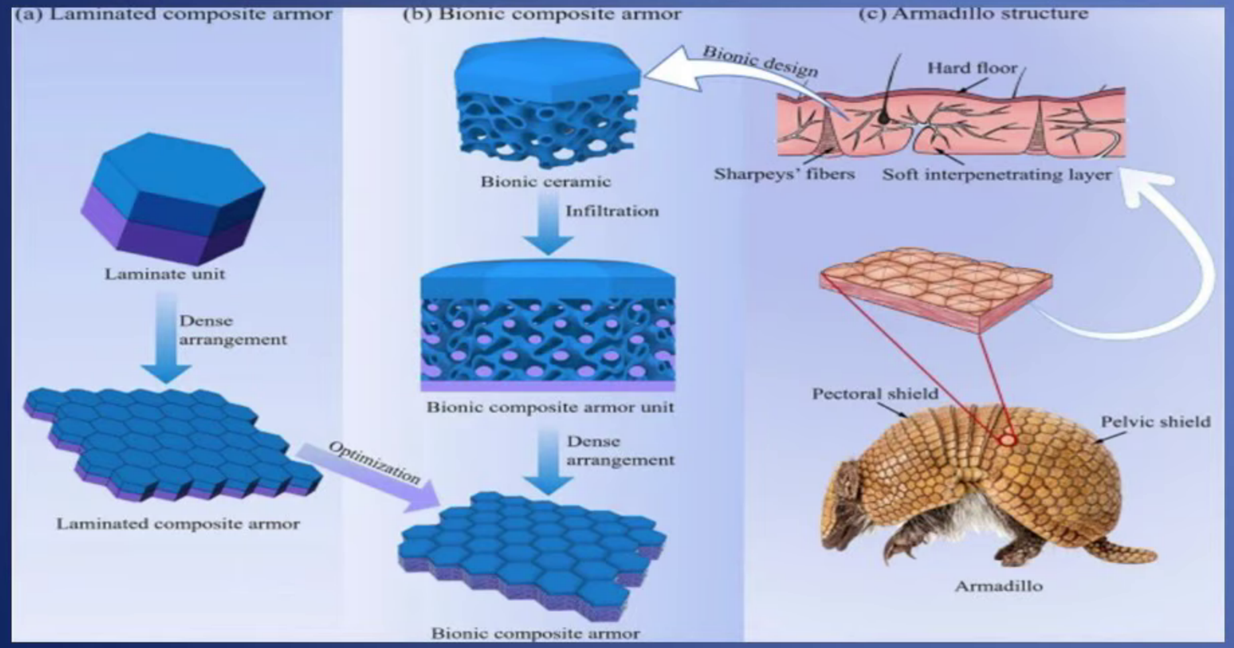
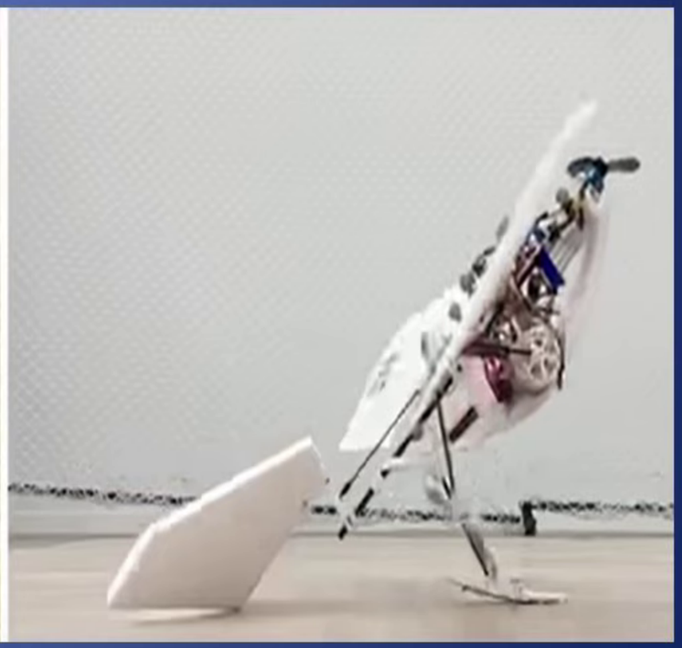
Evolution of Bio-Inspired Design

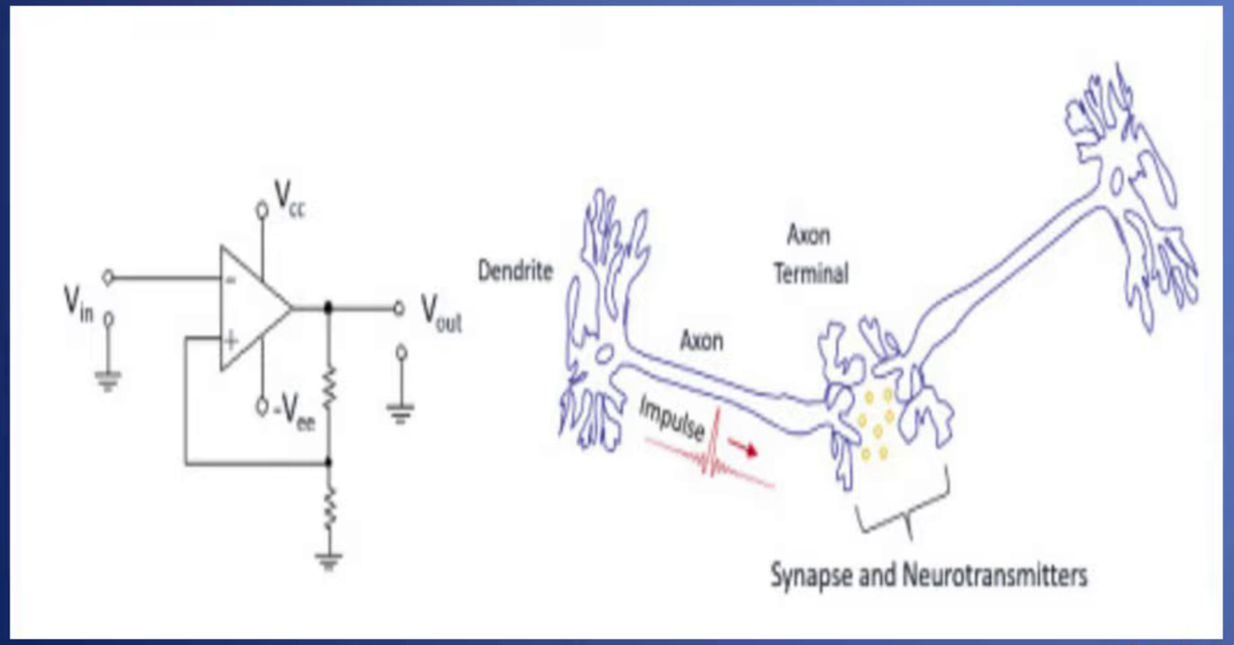
1. <u>Long before 'Bio-mimetics' was a word:</u>
   (a) Tortoise and Armadillo shells inspired Military Armors
   (b) Flying Machines inspired by birds





Evolution of Bio-Inspired Design

2. <u>Mid-20th Century (1950s):</u>
   (a) Otto H. Schmitt copied nerve behaviour for Schmitt trigger.
   (b) George de Mestral copied burr hooks for mechanized hook-and-loop fastener, later termed as Velcro.



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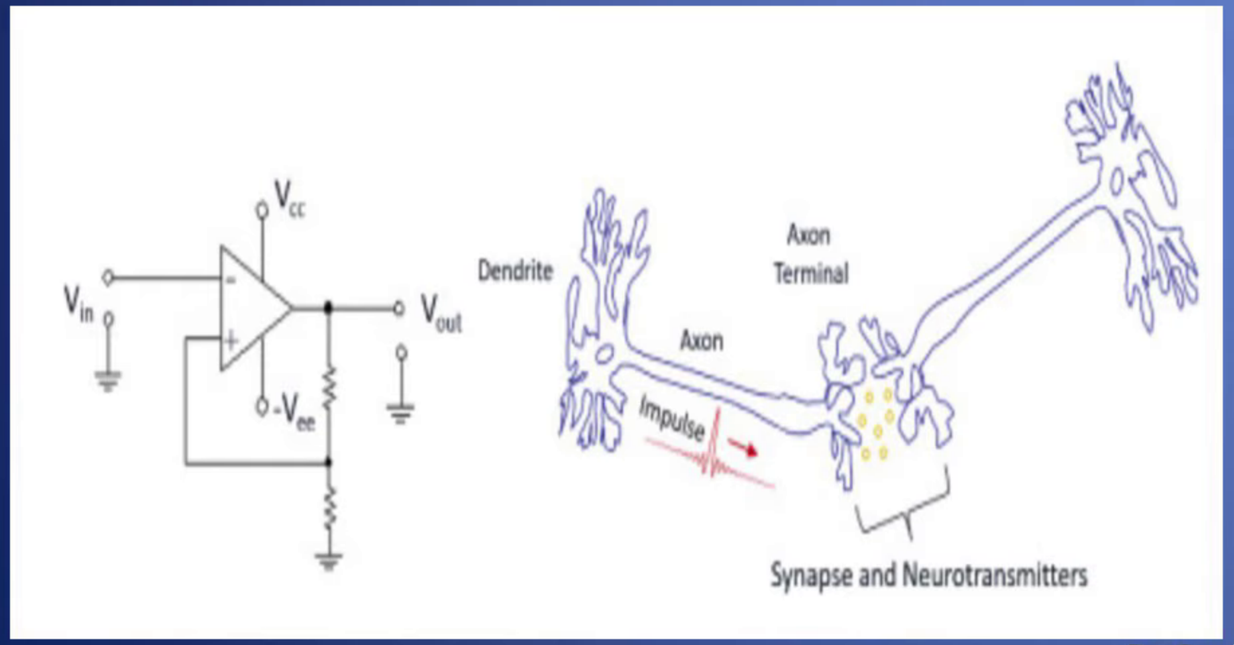

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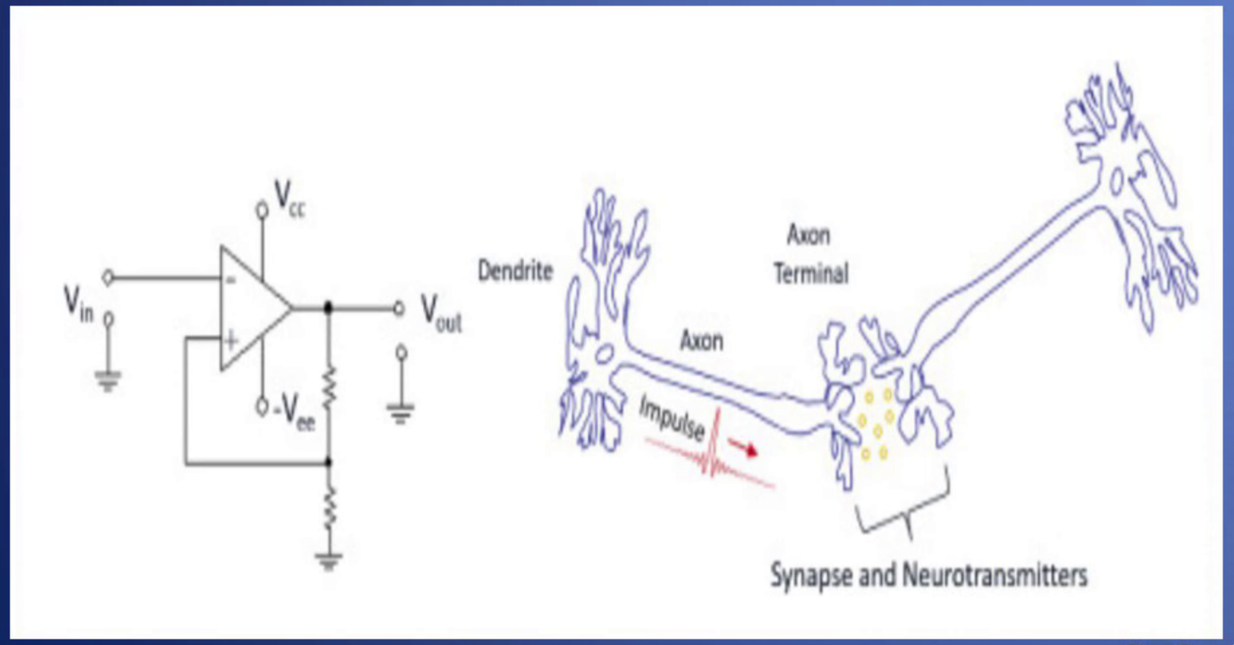
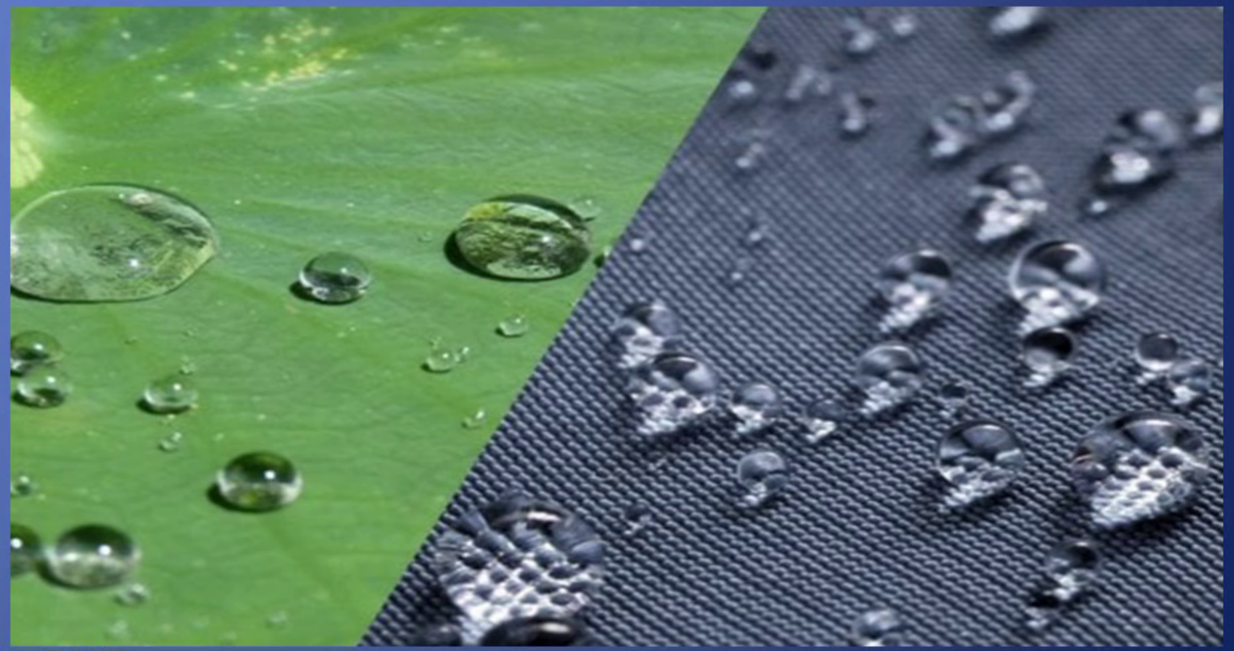


Evolution of Bio-Inspired Design

3. Late-20th Century(1990s–2000s) – Classic Biomimicry systems:
   (a) Termite mound inspired buildings: passive ventilation concept
   (b) Lotus-effect self-cleaning: biomimetic materials used for commercial coatings.



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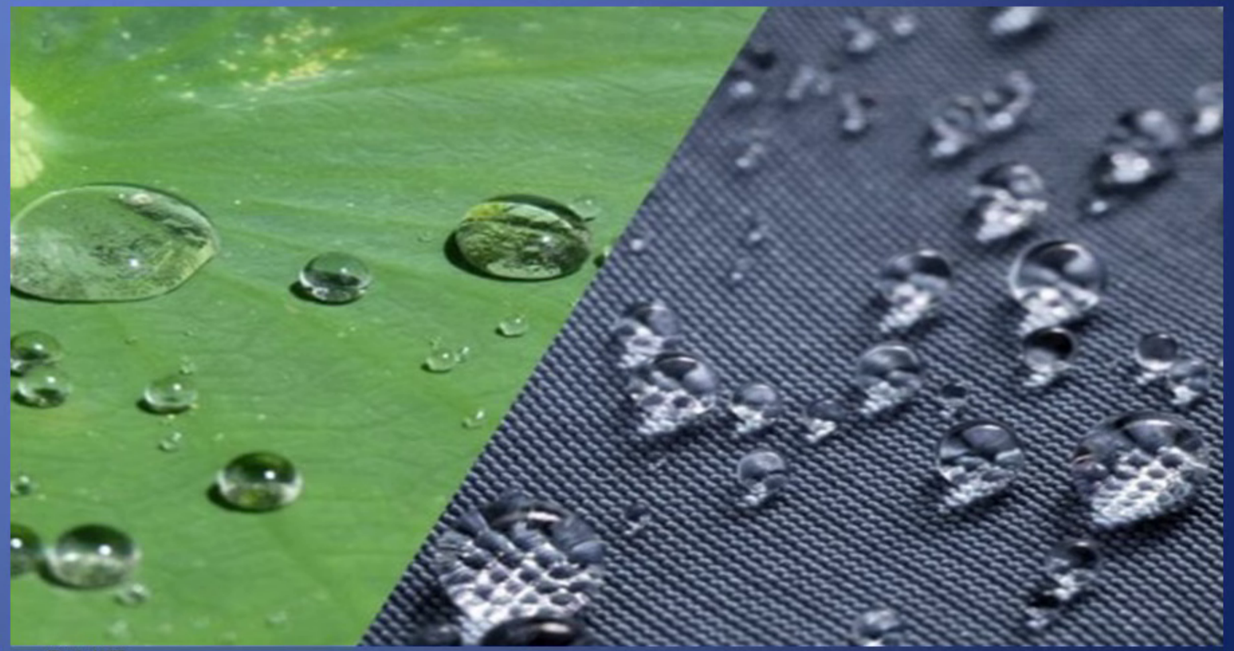
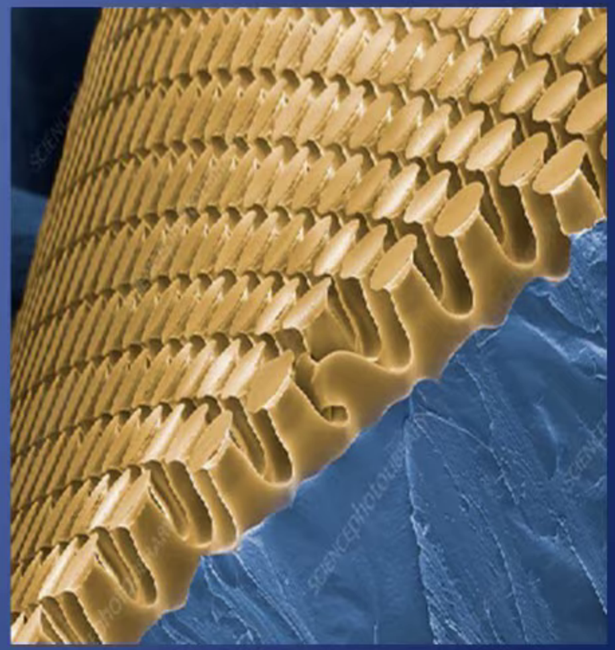


Evolution of Bio-Inspired Design

4. 21st Century – Inter-disciplinary systems:
   (a) Kingfisher inspired Shinkansen train nose.
   (b) Gecko setae inspired adhesives.



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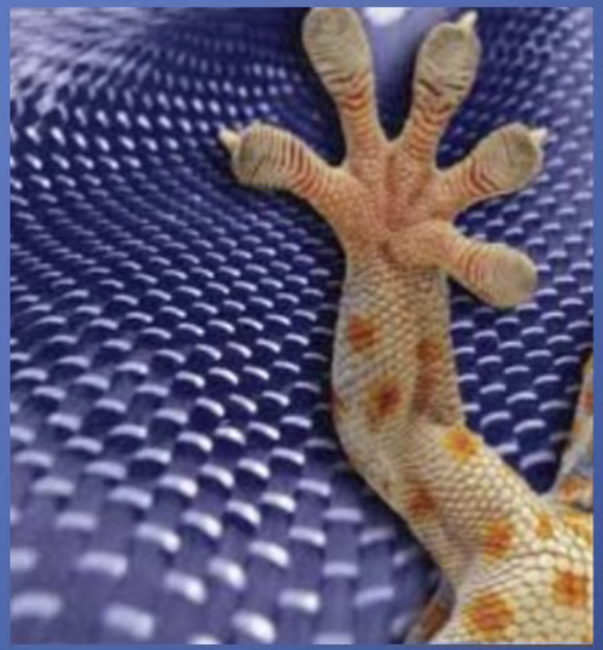

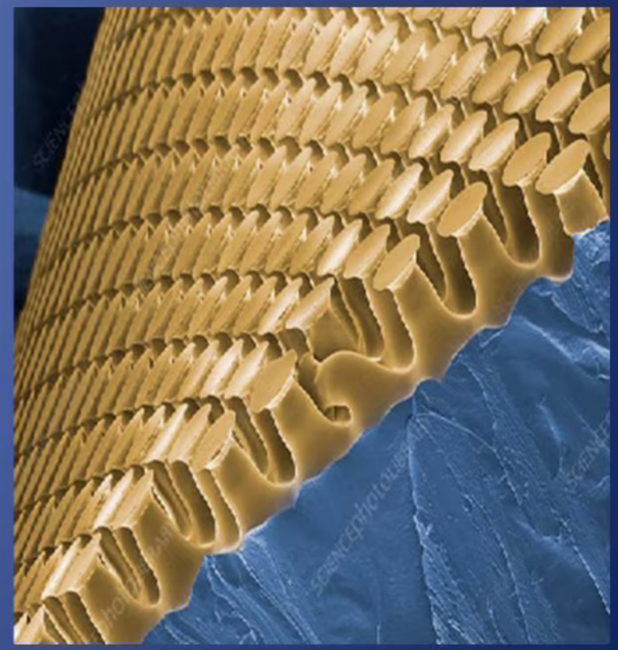

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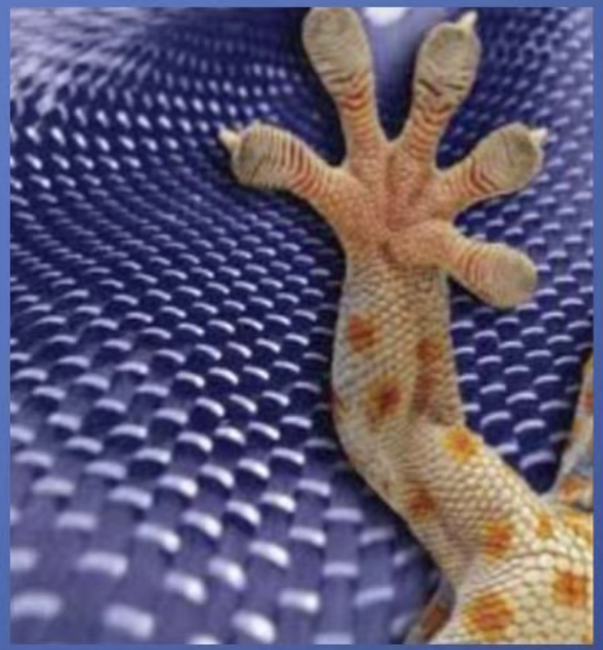

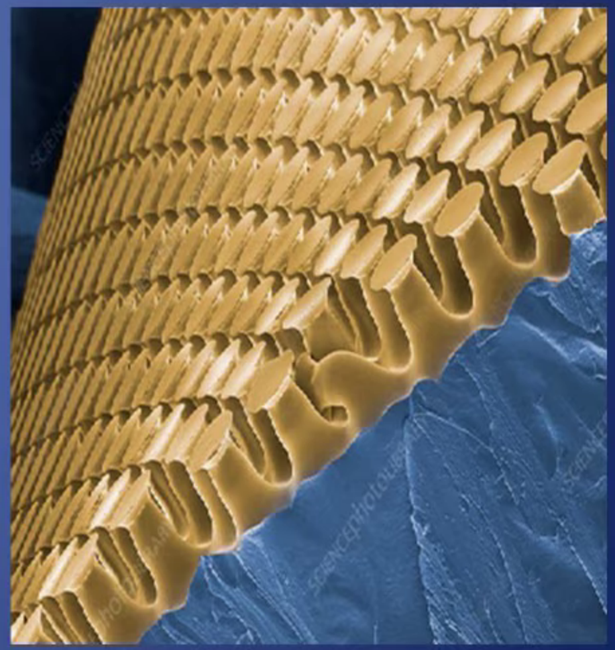

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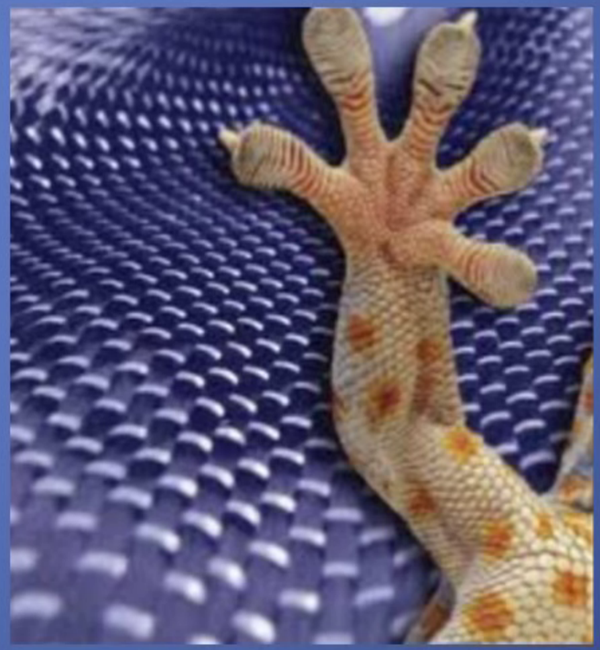

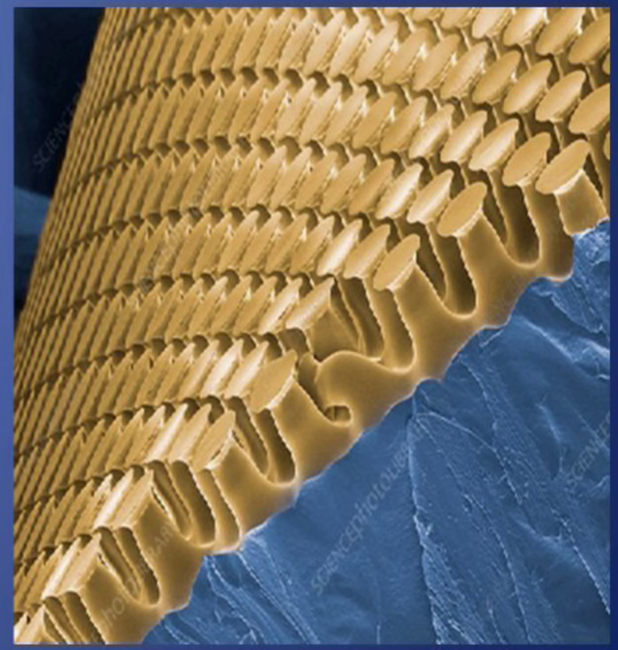

Web Resources for Bio-Inspired Design

S.No	Online Resource	Key Features
1	AskNature	Biological strategies database categorized by function; lots of case studies & inspiration. **URL:** https://biomimicry.org/inspiration/asknature/
2	The Biomimicry Institute	Core site for biomimicry: theory, methodology, learning tools, events. **URL:** https://biomimicry.org/innovation/biomimicry-co-labs/
3	Biomimicry 3.8	Professional training, tools, case studies, innovation consulting in nature-inspired design. **URL:** https://www.biomimicry.net/work-examples/

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3	Biomimicry 3.8	Professional training, tools, case studies, innovation consulting in nature-inspired design. **URL:** https://www.biomimicry.net/work-examples/

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8	Jacquelyn K. Nagel's Bio-Inspired Design Resources	BID Canvas, E2B Thesaurus, biological functional modeling. URL: https://jacquelynnagel.com/bid-resources/
9	NSF Workshop Report: Bio-Inspired Design	2022/2023 Detailed report on methods, applications, challenges in bio-inspired design. URL: https://www.nsf.gov/
10	TeachEngineering	Activities for introducing biomimicry. **URL:** https://www.teachengineering.org/activities/view/cub_bio_lesson05_activity1

Bio-Inspired Design

Design Forms:

- **Hexagonal Unit Cells**

  Example:

  Honey Combs by Bees.

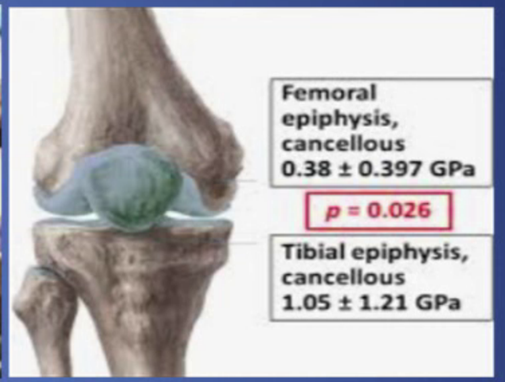
  Application:

  Efficient packing and high strength-to-weight ratio.

  It is used in aerospace panels and lightweight structures.

Bio-Inspired Design

Design Forms:



- **Anisotropy**

  Example:

  Bones, Woods, Tendons

  Application:

  Have different strengths along different axes.

  It is used in composite materials and adaptive load-bearing designs.

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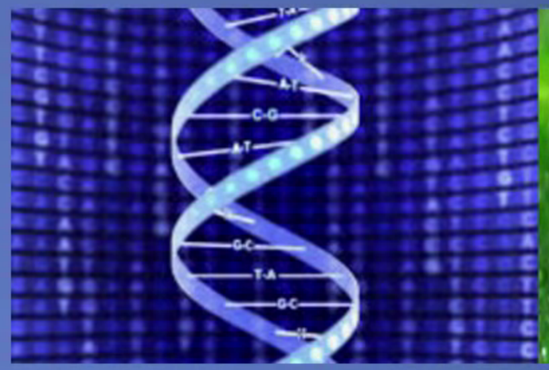

- **Helical & Spiral forms**

  Example:

  DNA helix, Snail Shells

  Application:

  Have compact storage and efficient growth patterns.

  It is used in springs, turbines and compact data storage designs.

Bio-Inspired Design

Design Forms:

- **Voronoi patterns**

  Example:

  Giraff's skin coat, Dragonfly wings

  Application:

  Naturally partition space with minimal material.

  It is used in crash-resistant car frame designs and lightweight architectures.

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VIT

CHENNAI